Spectral Leakage, Aliasing, and Jacobian Isotropy in S2 Activation for SO(3)-Equivariant Networks

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Abstract

S2 Activation—the standard nonlinearity in SO(3)-equivariant networks—applies a pointwise activation on the sphere via spherical harmonic (SH) expansion, truncation to $\ell_{\max}$, and finite quadrature. We present a systematic numerical study decomposing its errors along two orthogonal axes: *activation choice* and *quadrature strategy*. We identify a fundamental *truncation–aliasing dichotomy*: coefficient error is dominated by truncation (insensitive to oversampling), while equivariance error is dominated by aliasing (reduced 10–450× by 3× oversampling). Both axes are governed by spectral leakage, which is strongly correlated with equivariance violation. We further introduce a Jacobian-based isotropy metric to characterize how uniformly each activation explores output coefficient space, finding that smooth non-saturating activations (Softplus(1), SiLU) achieve both low leakage and high isotropy under $\mathrm{GL}_{1\times}$ quadrature. Preliminary task experiments show that neither leakage nor isotropy alone predicts downstream accuracy, highlighting the multi-factorial nature of activation selection.

1. Introduction

S2 Activation is the standard pointwise nonlinearity in SO(3)-equivariant neural networks, including SCN (Zitnick et al., 2022), eSCN (Passaro and Zitnick, 2023), and EquiformerV2 (Liao et al., 2024). Given spherical harmonic (SH) coefficients $\{c_{\ell m}\}$ for $\ell = 0, \ldots, \ell_{\max}$, the operator projects onto the sphere, applies a scalar nonlinearity σ, and integrates back via quadrature. Both the finite bandwidth $\ell_{\max}$ and the finite quadrature introduce error and can break SO(3) equivariance.

Prior work has studied specific quadrature schemes for S2 Activation (Geiger and Smidt, 2022) and observed that equivariance degrades with non-smooth activations, but a systematic decomposition of the error sources has been lacking. In this paper we:

1. Show that **spectral leakage** (energy generated above $\ell_{\max}$) is strongly correlated with equivariance error across activations (§3).
2. Identify a **truncation–aliasing dichotomy**: coefficient error is truncation-dominated (sampling-insensitive), while equivariance error is aliasing-dominated (sampling-sensitive) (§4).
3. Introduce a **Jacobian isotropy** metric that measures how uniformly S2 Activation explores output directions, and show it is orthogonal to leakage (§5).

2. Background and Definitions

S2 Activation pipeline. Let $f : S^2 \to \mathbb{R}$ be a band-limited function with SH coefficients $\{c_{\ell m}\}$ for $\ell \le \ell_{\max}$, $m \in \{-\ell, \ldots, \ell\}$. The S2 Activation A_σ proceeds in three steps:

$$(1) \quad f(x_i) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m}\, Y_{\ell m}(x_i), \tag{1}$$

$$(2) \quad g(x_i) = \sigma\big(f(x_i)\big), \tag{2}$$

$$(3) \quad c'_{\ell m} = \sum_{i=1}^{N} w_i\, g(x_i)\, Y_{\ell m}(x_i), \tag{3}$$

where $\{x_i, w_i\}_{i=1}^N$ is a quadrature rule on S^2 with $\sum_i w_i = 4\pi$.

Angular power spectrum. $P(\ell) = \sum_{m=-\ell}^{\ell} |c_{\ell m}|^2$ measures the total energy at degree ℓ.

Spectral leakage ratio. After applying σ to a band-limited input ($\ell \le \ell_{\max}$), the output generally has energy at all degrees. We measure the leakage via a high-resolution reference quadrature ($\ell_{\max}^{\mathrm{ref}} = 25$):

$$R = \frac{\sum_{\ell > \ell_{\max}} P_g(\ell)}{\sum_{\ell \ge 0} P_g(\ell)}, \tag{4}$$

where P_g is the power spectrum of $g = \sigma \circ f$.

Coefficient error. $E_{\mathrm{coeff}} = \|c'_{\mathrm{out}} - c'_{\mathrm{gt}}\| / \|c'_{\mathrm{gt}}\|$, where c'_{gt} is obtained with GL quadrature at $\ell_{\max}^{\mathrm{ref}} = 25$, truncated to $\ell_{\max}$.

Equivariance error. For rotation $R \in \mathrm{SO}(3)$ with Wigner-D matrix $D(R)$:

$$E_{\mathrm{equiv}} = \frac{\|A_\sigma(D(R)\, c) - D(R)\, A_\sigma(c)\|}{\|A_\sigma(D(R)\, c)\|}, \tag{5}$$

averaged over 20 random rotations and 10 random inputs.

Experimental setup. Inputs: $c_{\ell m} \sim \mathcal{N}(0,1)$ i.i.d., giving $(\ell_{\max}+1)^2$ coefficients. We test $\ell_{\max} \in \{3,6,10\}$ and focus on $\ell_{\max} = 6$ ($d = 49$). Activations: C^∞ smooth (Softplus(β=1), SiLU, GELU, tanh), C^0 (ReLU, $|\cdot|$), and the parametric family Softplus(β) with $\beta \in \{1,3,10,30,100\}$. Quadrature: Gauss–Legendre (GL), Lebedev, and uniform (equiangular) grids.

3. Equivariance Error vs. Activation Function

We first investigate how the activation choice affects equivariance, holding the quadrature fixed at $\mathrm{GL}_{1\times}$.

Spectral leakage across activations. Figure 1 shows the leakage ratio R for each activation at $\ell_{\max} = 3, 6, 10$. The ranking is consistent: C^∞ non-saturating activations (Softplus(1): R=0.035; SiLU: R=0.108) have lower leakage than C^0 activations (ReLU: R=0.090; $|\cdot|$: R=0.160). The Softplus(β) family interpolates continuously: R increases monotonically with β, confirming that leakage is controlled by activation smoothness. Notably, tanh (R=0.094) has moderate leakage despite being C^∞, because saturation compresses the output range.

Leakage predicts equivariance error. Figure 2 plots E_{equiv} against R for each activation. The strong positive correlation confirms the mechanistic link: leaked energy aliases back into $\ell \le \ell_{\max}$ coefficients in a rotation-dependent manner, since different rotations move the function relative to the fixed quadrature grid, producing different aliasing patterns.

4. Equivariance Error vs. Sampling Strategy

4.1 The Truncation–Aliasing Dichotomy

Coefficient error is sampling-insensitive. Figure 3 shows E_{coeff} vs. GL oversampling ratio (1×, 2×, 3×). The curves are nearly flat: increasing quadrature points from ∼100 to ∼700 barely changes the output coefficients. This is because the error is dominated by *truncation*: when σ generates energy at $\ell > \ell_{\max}$, that energy is lost regardless of quadrature accuracy—the output SH basis simply cannot represent it.

Equivariance error is sampling-sensitive. In contrast, Figure 4 shows that $\mathrm{GL}_{3\times}$ reduces E_{equiv} by 10–450× compared to $\mathrm{GL}_{1\times}$, depending on the activation (Table 1).

Why the dichotomy? The key insight is that *truncation error is equivariant but aliasing error is not*. Truncation is a projection onto the $\ell \le \ell_{\max}$ subspace, which commutes with SO(3) rotations (each degree ℓ transforms independently under the Wigner-D matrices). It loses information but loses the *same* information regardless of orientation. Aliasing, however, arises when the quadrature cannot exactly integrate products of high-ℓ and low-ℓ harmonics. The fold-back pattern depends on the function's orientation relative to the fixed quadrature grid, so it changes with rotation, breaking equivariance. Oversampling reduces aliasing by providing enough quadrature points to accurately integrate these cross-terms. Smoother activations benefit more because they have less high-ℓ content to alias.

Table 1: Equivariance error under GL oversampling ($\ell_{\max}$=6).

Activation	$\mathrm{GL}_{1\times}$	$\mathrm{GL}_{3\times}$	Ratio
Softplus(1)	0.142	3.2×10^{-4}	450×
SiLU	0.260	1.6×10^{-3}	162×
ReLU	0.244	1.1×10^{-2}	23×
tanh	0.281	1.3×10^{-2}	21×
$\lvert\cdot\rvert$	0.333	1.5×10^{-2}	22×

4.2 Comparison of Quadrature Methods

Figure 5 compares GL, Lebedev, and uniform grids at $\ell_{\max}$=6. GL slightly outperforms Lebedev at matched oversampling ($\mathrm{GL}_{1\times}$ vs. Leb-min), likely due to optimal placement for the tensor-product structure of S^2. Lebedev at 2× degree (230 points) performs comparably to $\mathrm{GL}_{2\times}$ (338 points) for smooth activations. Uniform grids at high resolution (N=5000) achieve the lowest error across all activations. The relative ranking of activations is preserved across all quadrature methods, confirming that the smoothness effect is independent of quadrature choice.

5. Jacobian Isotropy Analysis

Equivariance is not the only property that matters for downstream tasks: the activation must also produce diverse nonlinear transformations on the sphere. We quantify this via the Jacobian of the (discretized) S2 Activation operator.

Definition. The S2 Activation $A_\sigma : \mathbb{R}^d \to \mathbb{R}^d$ maps input SH coefficients c to output coefficients $c' = A_\sigma(c)$. Its Jacobian at c is $J(c) = \partial A_\sigma / \partial c$, a $d\times d$ matrix with singular values $s_1 \ge s_2 \ge \cdots \ge s_d$. We define the *Jacobian isotropy* as the *effective rank*:

$$\mathrm{EffRank}(J) = \exp\Big(-\sum_i p_i \log p_i\Big), \quad p_i = \frac{s_i^2}{\sum_j s_j^2}, \tag{6}$$

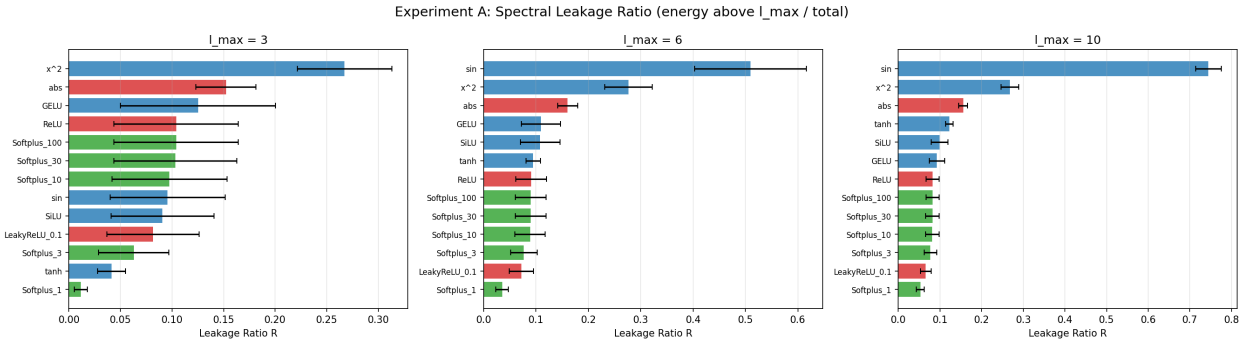

Figure 1: Spectral leakage ratio R (Eq. 4) for each activation at $\ell_{\max} = 3, 6, 10$. Red: C^0. Green: Softplus family. Blue: C^∞. Error bars: std over 10 random inputs.

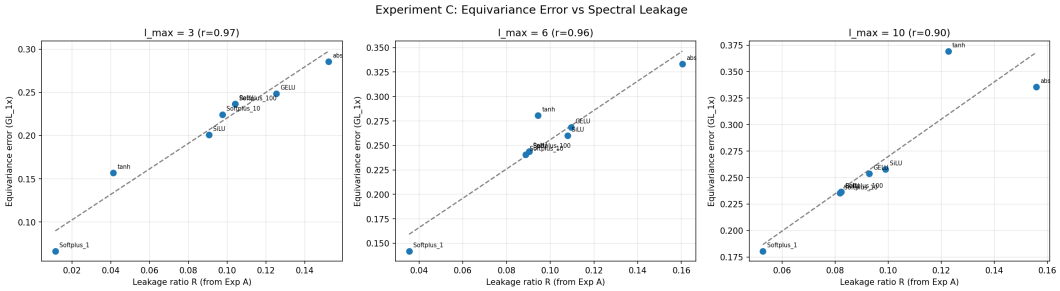

Figure 2: Equivariance error E_{equiv} vs. leakage ratio R at $\ell_{\max} = 3, 6, 10$ ($\text{GL}_{1\times}$). Each point is one activation. The correlation shows leakage is a strong predictor of equivariance violation.

which counts how many directions in output space the operator locally explores. $\text{EffRank} = d$ means perfectly isotropic (all s_i equal); $\text{EffRank} = 1$ means all output variation is along a single direction.

Important caveat: Effective rank measures the *distribution* of singular values, not their magnitude. A Jacobian εI has maximal effective rank but negligible output variation. We therefore report effective rank alongside the Frobenius norm $\|J\|_F$ in the supplementary material. For all activations tested, $\|J\|_F$ is of similar order, so effective rank is a meaningful comparison.

Computation. From the S2 Activation forward pass (1)–(3), the Jacobian is $J = W_Y^\top \operatorname{diag}(\sigma'(Yc))\, Y$, where Y is the SH evaluation matrix and W_Y the weighted inverse matrix. We compute J via autograd and take its SVD. Results are averaged over 50 random inputs ($c_{\ell m} \sim \mathcal{N}(0, 1)$).

Note: The effective rank depends on the specific discretization ($\text{GL}_{1\times}$ here). All isotropy claims in this paper are scoped to this quadrature.

Results. Figure 6 shows the effective rank at $\ell_{\max} = 3, 6, 10$, revealing a pattern that contradicts naive intuitions:

- **Softplus(1) has the highest effective rank** (33.5/49 at $\ell_{\max}$=6), because its smooth, non-saturating derivative $\sigma'(f)$ is nonzero and varies gradually across the sphere, producing a well-conditioned Jacobian.
- **tanh has low effective rank** (25.6/49), because saturation ($\tanh'(f) \to 0$ for large $|f|$) collapses many output directions.
- $|\cdot|$ **has the highest rank** (39.6/49), because $|\sigma'(f)| = 1$ a.e., preserving all directions despite being C^0.
- **Softplus(β) rank decreases with** β: as β increases toward ReLU, the derivative zeros out for $f < 0$, collapsing ∼half the directions.

Figure 7 plots isotropy against leakage, showing the two metrics are separable. Under $\text{GL}_{1\times}$, Softplus(1) and SiLU sit in the low-leakage / high-isotropy quadrant, while tanh has lower isotropy due to saturation and $|\cdot|$ has high isotropy but high leakage.

6. Pilot: Downstream Task Performance

We train a 1-layer SphericalCNN on a synthetic 5-class classification task ($\ell_{\max}$=6, class-discriminative signal in high-ℓ coefficients). 5 activations × 3 sampling configs, 2 runs each, 15 epochs.

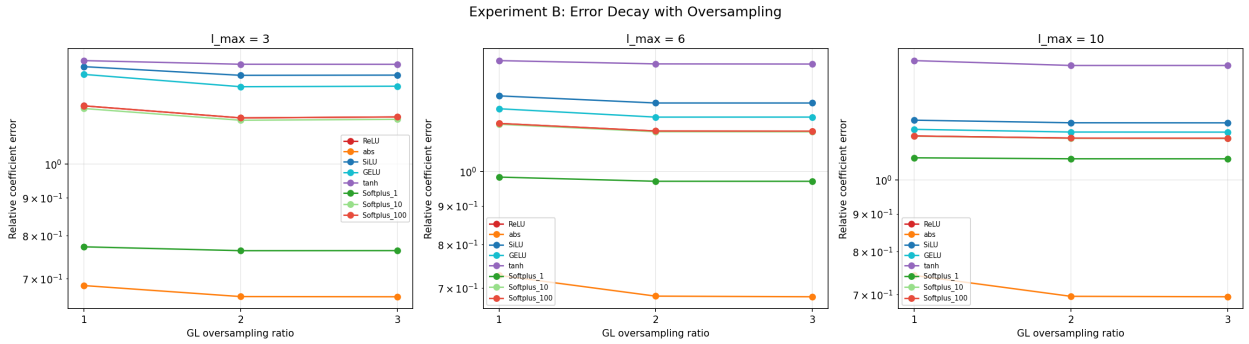

Figure 3: Coefficient error E_{coeff} vs. GL oversampling ratio. Near-flat curves show that quadrature resolution does not reduce coefficient error, because truncation dominates.

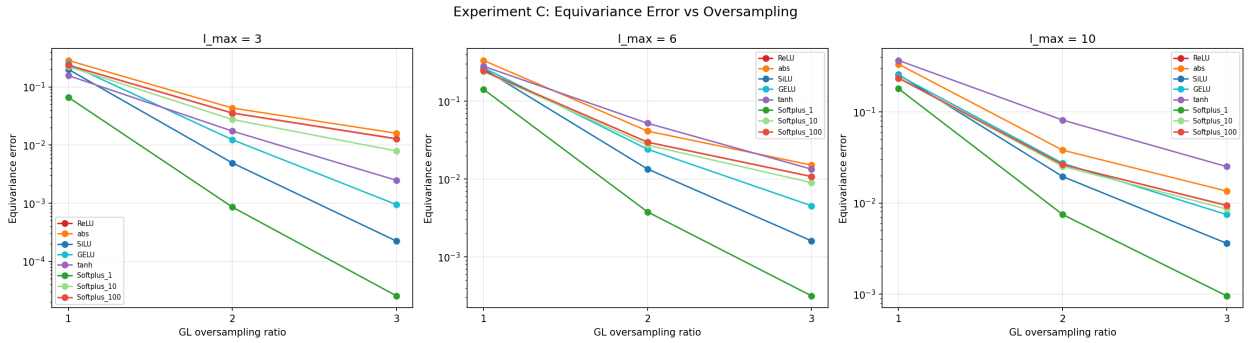

Figure 4: Equivariance error E_{equiv} vs. GL oversampling ratio. Compare with Fig. 3: while coefficient error is flat, equivariance error drops 1–2 orders of magnitude with $3\times$ oversampling.

Results. Accuracy differences are small ($\sim$1–2%): tanh 87.5%, Softplus(1) 87.3%, SiLU 86.3%, ReLU 86.3%, Softplus(10) 86.0% ($\text{GL}_{1\times}$). Neither leakage nor Jacobian isotropy alone predicts this ranking. In particular, tanh achieves the highest accuracy despite having *low* isotropy and *moderate* leakage, while Softplus(1)—which is Pareto-optimal on both axes—is a close second. With only n=2 runs per configuration and $\sim$1% accuracy gaps, these differences are not statistically significant; we present them as pilot evidence only.

The disconnect suggests that for a 1-layer model, the global shape of the nonlinearity (saturation produces bounded, well-separated class responses) may matter more than local Jacobian diversity. In deeper architectures where equivariance errors compound across layers, the activation ranking could shift.

7. Conclusions

We identify three orthogonal axes characterizing S2 Activation:

1. Smoothness $\rightarrow$ leakage $\rightarrow$ equivariance. Smoother activations produce less spectral leakage, which is strongly correlated with equivariance error. The Softplus(β) family provides continuous parametric control.

2. Truncation–aliasing dichotomy. Coefficient error is truncation-dominated (sampling-insensitive); equivariance error is aliasing-dominated (sampling-sensitive). This is because truncation commutes with rotation while aliasing does not.

3. Isotropy is orthogonal to leakage. Under $\text{GL}_{1\times}$ discretization, the Jacobian effective rank reveals that smooth non-saturating activations (Softplus(1), SiLU) are also the most isotropic, while saturating (tanh) or half-zeroing (ReLU) activations lose directions. This provides a second axis for characterizing activations beyond smoothness alone.

4. Task performance remains open. Pilot experiments show that neither leakage nor isotropy alone predicts downstream accuracy. Controlled studies matching model capacity across activations, and evaluation on deeper architectures, are needed.

Practical guidance (for equivariance). Softplus(1) or SiLU with $\geq 2\times$ GL oversampling minimizes equivariance error. The cost-effective default is SiLU + $\text{GL}_{1\times}$. Minimizing equivariance error does not guarantee best

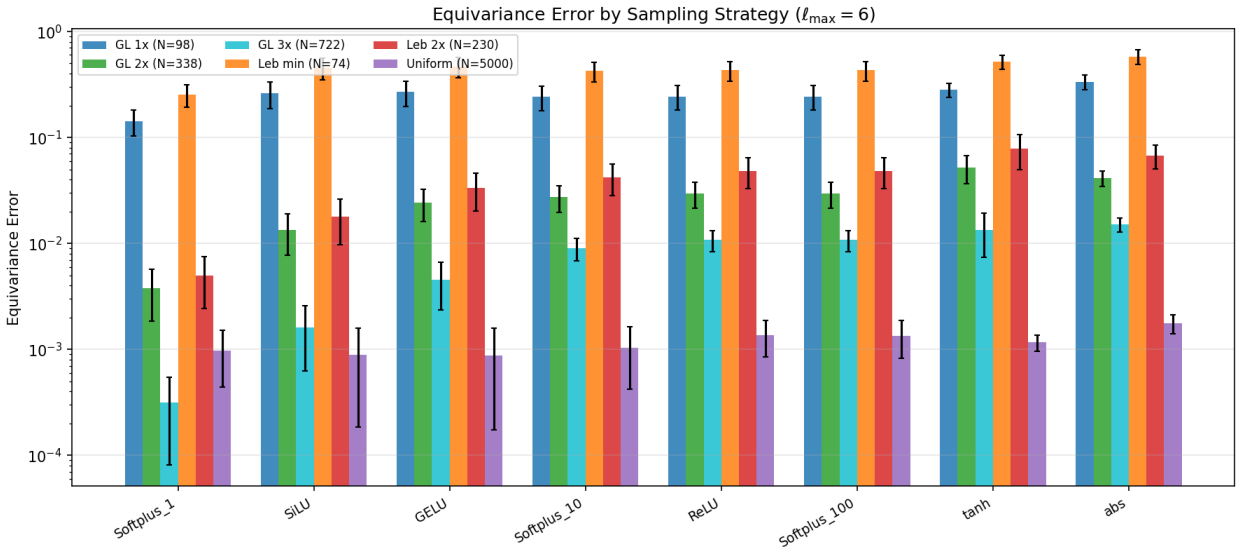

Figure 5: E_{equiv} under six sampling strategies at $\ell_{\max}$=6. N: number of quadrature points. Log scale. The activation ranking is stable across methods.

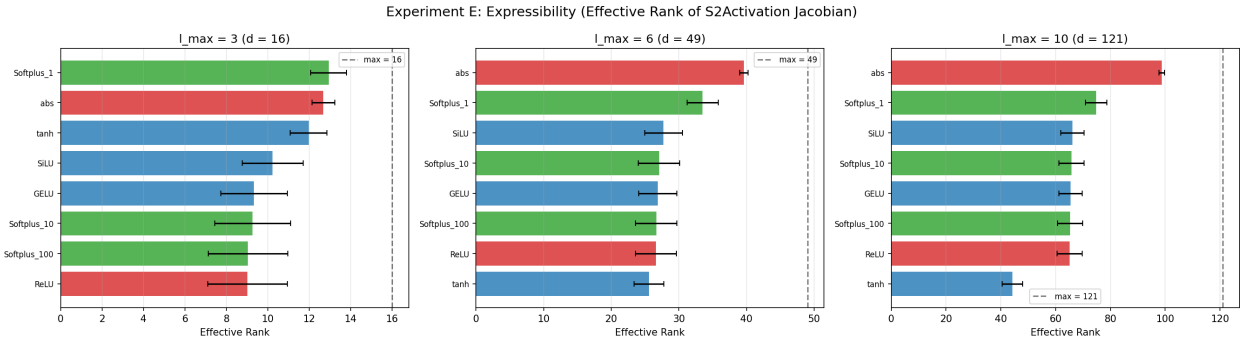

Figure 6: Effective rank (Eq. 6) of the S2 Activation Jacobian for $\text{GL}_{1\times}$ at $\ell_{\max} = 3, 6, 10$. Dashed: max rank d. Averaged over 50 random inputs.

task performance in shallow models.

References

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A. Additional Figures

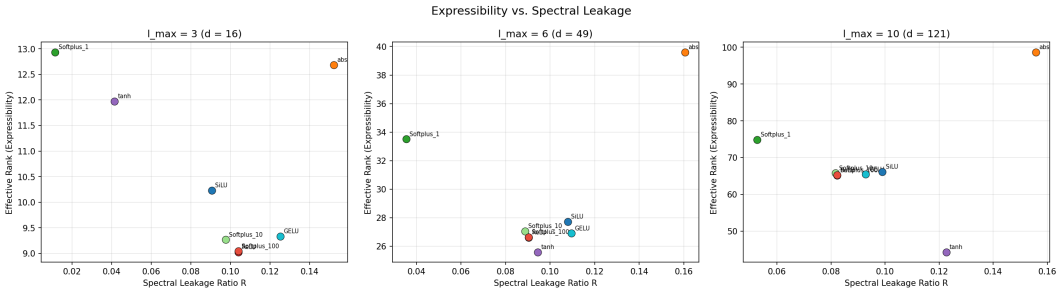

Figure 7: Effective rank vs. spectral leakage at $\ell_{\max} = 3, 6, 10$ ($\mathrm{GL}_{1\times}$). The two axes measure different properties: leakage quantifies energy lost above $\ell_{\max}$; isotropy quantifies output diversity within $\ell_{\max}$.

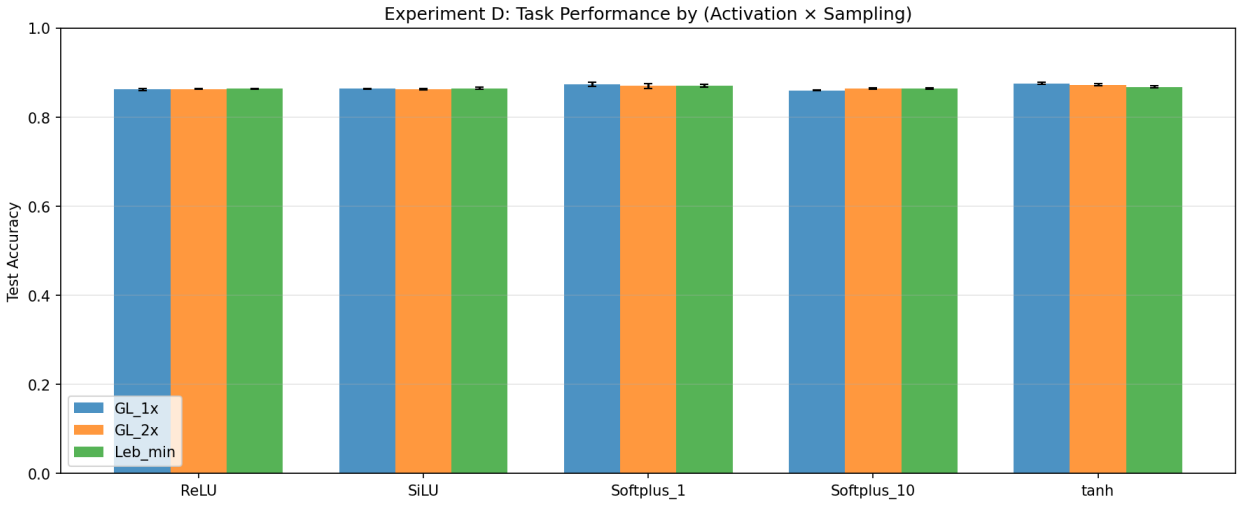

Figure 8: Test accuracy by (activation $\times$ sampling) configuration. 2 runs per config, 15 epochs, $\ell_{\max}$=6.

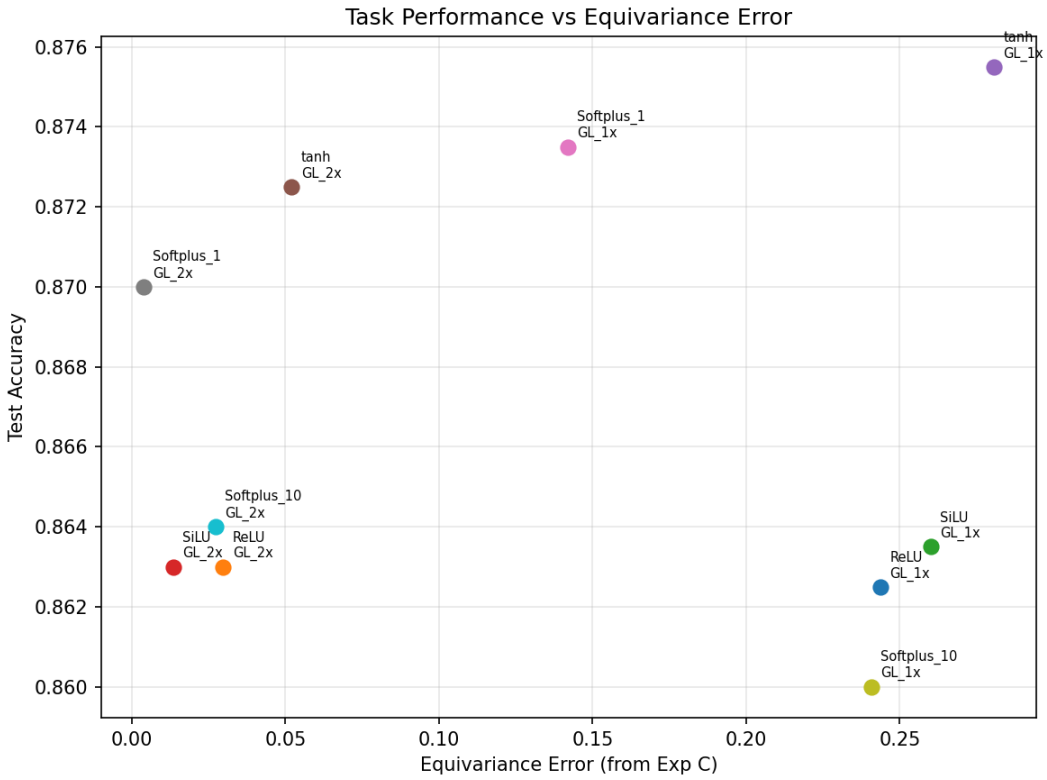

Figure 9: Task accuracy vs. equivariance error. No clear monotonic trend.